Experiment 05:

Laboratory Exercise:

Part A: Identifying and performing statistical analysis

1.Mean

Code:

```
mean_values=df.mean(numeric_only=True)
print("\n--------------------MEAN--------------------\n")
print(mean_values)
```

Output:

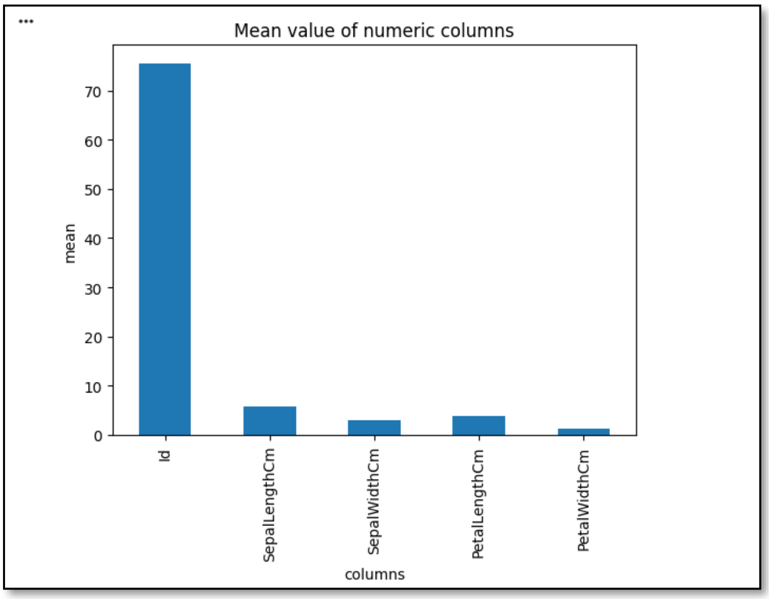
```
--------------------MEAN--------------------

Id               75.500000
SepalLengthCm     5.843333
SepalWidthCm      3.054000
PetalLengthCm     3.758667
PetalWidthCm      1.198667
dtype: float64
```

Code:

```
mean_values.plot(kind='bar')
plt.title('Mean value of numeric columns')
plt.xlabel('columns')
plt.ylabel('mean')
plt.show()
```

Output:



Observation:

- The mean **Sepal Length** is approximately **5.84** cm, while the mean Sepal Width is about **3.05** cm.
- The mean Petal Length is approximately **3.76** cm and the mean **Petal Width** is about **1.20** cm.
- Among the four measurements, **Sepal Length** has the highest mean value.

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2. Variance & Standard Deviation:

Code:

```
variance=df.var(numeric_only=True)
std_dev=df.std(numeric_only=True)
print("\n--------------------VARIANCE--------------------\n")
print(variance)
print("\n---------------------STANDARD-DEVIATION---------------------\n")
print(std_dev)
```

Output:

```
...
    ---------------------VARIANCE--------------------

    Id                1887.500000
    SepalLengthCm        0.685694
    SepalWidthCm         0.188004
    PetalLengthCm        3.113179
    PetalWidthCm         0.582414
    dtype: float64

    ---------------------STANDARD-DEVIATION--------------------

    Id                43.445368
    SepalLengthCm      0.828066
    SepalWidthCm       0.433594
    PetalLengthCm      1.764420
    PetalWidthCm       0.763161
    dtype: float64
```

Observation:

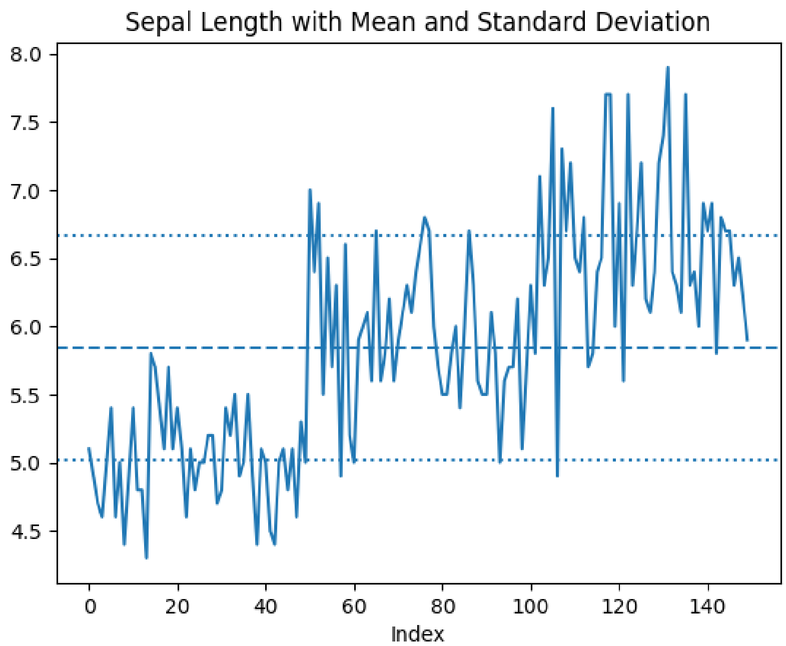
- Sepal Length has a variance of approximately **0.69**, while Sepal Width has a variance of about **0.19**.
- Petal Length has the **highest variance of about 3.11**, showing greater variation in its values.
- The standard deviation of Petal Length is also the highest, at approximately **1.76**.

Code:

```
sepal_length=df['SepalLengthCm']

print('sepal length data : ',sepal_length.head())

mean_sepal=sepal_length.mean()
var_sepal=sepal_length.var()
std_sepal=sepal_length.std()

# Plotting
plt.figure()

# Plot actual sepal length values
plt.plot(sepal_length.values, label='Sepal Length')

# Plot mean line
plt.axhline(mean_sepal, linestyle='--', label='Mean')

# Plot standard deviation lines
plt.axhline(mean_sepal + std_sepal, linestyle=':', label='Mean + Std Dev')
plt.axhline(mean_sepal - std_sepal, linestyle=':', label='Mean - Std Dev')

plt.title("Sepal Length with Mean and Standard Deviation")
plt.xlabel("Index")
plt.show()
print('variance of sepal length :',var_sepal)
```

Output:

```
sepal length data :  0    5.1
1    4.9
2    4.7
3    4.6
4    5.0
Name: SepalLengthCm, dtype: float64
```




```
variance of sepal length : 0.6856935123042505
```

Observation:

- The **Sepal Length** values vary around the mean value of approximately **5.84**.
- Most of the values are seen around the mean and within the standard deviation range.
- The graph shows noticeable variation in **Sepal Length** values, especially in the later part of the dataset.

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3. Correlation:

Code:

```
correlation=df['SepalWidthCm'].corr(df['PetalLengthCm'])
print('correlation :',correlation)

correlation=df['PetalWidthCm'].corr(df['PetalLengthCm'])
print('correlation :',correlation)

correlation=df['PetalWidthCm'].corr(df['PetalWidthCm'])
print('correlation :',correlation)
```

Output:

```
correlation : -0.42051609640115445
correlation : 0.9627570970509663
correlation : 1.0
```

Observation:

- Sepal Width and Petal Length show a negative correlation of about **-0.42**.
- **Petal Width** and Petal Length show a strong positive correlation of about **0.96**.
- **Petal Width** with itself has a correlation of **1**, showing a perfect positive relationship.

---

4. Covariance:

Code:

```
covariance=df['SepalWidthCm'].cov(df['PetalLengthCm'])
print('covariance :',covariance)

covariance=df['PetalWidthCm'].cov(df['PetalLengthCm'])
print('covariance :',covariance)

covariance=df['PetalWidthCm'].cov(df['PetalWidthCm'])
print('covariance :',covariance)
```

Output:

```
•••  covariance : -0.32171275167785235
     covariance : 1.2963874720357944
     covariance : 0.5824143176733781
```

Observation:

- Sepal Width and Petal Length have a negative covariance of about **-0.32**.
- **Petal Width** and Petal Length have a positive covariance of about **1.30**.
- **Petal Width** with itself has a positive covariance of approximately **0.58**.

---

5. Mean Matrix:

Code:

```
numerical_cols=['SepalLengthCm','SepalWidthCm','PetalLengthCm','PetalWidthCm']
df[numerical_cols].mean()
```

Output:

	0
SepalLengthCm	5.843333
SepalWidthCm	3.054000
PetalLengthCm	3.758667
PetalWidthCm	1.198667

Observation:

- The mean values of the four numerical attributes are displayed in the matrix.
- **Sepal Length** has the highest mean value of approximately **5.84**.
- **Petal Width** has the lowest mean value of approximately **1.20**.

---

6. Variance Matrix:

Code:

```
df[numerical_cols].var()
```

Output:

	0
SepalLengthCm	0.685694
SepalWidthCm	0.188004
PetalLengthCm	3.113179
PetalWidthCm	0.582414

dtype: float64

Observation:

- The variance values show different levels of spread among the four attributes.
- Petal Length has the highest variance, approximately **3.11**.
- Sepal Width has the lowest variance, approximately **0.19**, indicating comparatively less variation.

---

7. Standard Deviation Matrix:

Code:

```
df[numerical_cols].std()
```

Output:

	0
SepalLengthCm	0.828066
SepalWidthCm	0.433594
PetalLengthCm	1.764420
PetalWidthCm	0.763161

dtype: float64

Observation:

- Petal Length has the highest standard deviation of approximately **1.76**.
- Sepal Width has the lowest standard deviation, approximately **0.44**.
- This shows that Petal Length values are more spread out compared to the other attributes.

---

8. Correlation Matrix:

Code:

```
df[numerical_cols].corr()
```

Output:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
SepalLengthCm	1.000000	-0.109369	0.871754	0.817954
SepalWidthCm	-0.109369	1.000000	-0.420516	-0.356544
PetalLengthCm	0.871754	-0.420516	1.000000	0.962757
PetalWidthCm	0.817954	-0.356544	0.962757	1.000000

Observation:

- **Sepal Length** and Petal Length have a strong positive correlation of about **0.87**.
- Petal Length and **Petal Width** have the strongest positive correlation of about **0.96**.
- Sepal Width has a negative correlation with Petal Length and **Petal Width**.

---

9. Covariance Matrix:

Code:

```
df[numerical_cols].cov()
```

Output:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
SepalLengthCm	0.685694	-0.039268	1.273682	0.516904
SepalWidthCm	-0.039268	0.188004	-0.321713	-0.117981
PetalLengthCm	1.273682	-0.321713	3.113179	1.296387
PetalWidthCm	0.516904	-0.117981	1.296387	0.582414

Observation:

- **Sepal Length** and Petal Length have a positive covariance of about **1.27**.
- Sepal Width and Petal Length have a negative covariance of about **-0.32**.
- Petal Length and **Petal Width** have a positive covariance of about **1.30**, showing that they tend to increase together.

Part B: Applying hypothesis testing techniques

B.1: T-Test and its interpretation

Code:

```
from scipy import stats

sepal_length=df['SepalLengthCm']

mu_0=5.5
alpha=0.05

t_stat, p_value=stats.ttest_1samp(sepal_length,mu_0)
print("T-statistic :",t_stat)
print("P-value :",p_value)

if p_value<t_stat:
```

```
  print("\nDecision: Reject Null Hypothesis")
else:
  print("\nDecision: Fail to Reject Null Hypothesis")
```

Output:

```
...  T-statistic : 5.0780453996425345
     P-value : 1.1231119508340797e-06

     Decision: Reject Null Hypothesis
```

Observation:

- The calculated T-statistic is approximately **5.88** and the p-value is extremely small.
- The p-value is less than the significance level of **0.05**.
- Therefore, the output gives the decision to **Reject the Null Hypothesis**.

B.2 Z-Test

Code:

```
sepal_length=df['SepalLengthCm']

mu_0=5.5
sigma=0.8
alpha=0.05

sample_mean=sepal_length.mean()
sample_size=len(sepal_length)

z_stat=(sample_mean-mu_0)/(sigma/np.sqrt(sample_size))

critical_value=1.96

print("Z-statistic :",z_stat)
print("Critical Z-value :",critical_value)

if abs(z_stat)>critical_value:
  print("\nDecision: Reject Null Hypothesis")
else:
  print("\nDecision: Fail to Reject Null Hypothesis")
```

Output:

```
...  Z-statistic : 5.256196739722242
     Critical Z-value : 1.96

     Decision: Reject Null Hypothesis
```

Observation:

- The calculated Z-statistic is approximately **5.26**.
- The calculated value is greater than the critical Z-value of **1.96**.
- Therefore, the output gives the decision to **Reject the Null Hypothesis**.

Post-Experiments Exercise:

B. Questions:

1. Download a dataset containing numerical attributes. Calculate mean and variance for two attributes and write your observations.

Code:

```
# Load the Wine Quality dataset

wineData = pd.read_csv("/kaggle/input/wine-quality-dataset/WineQT.csv")
print(wineData.head())
print("----------------------------------------------------------------------")

# Calculate mean

alcoholMean = wineData["alcohol"].mean()
pHMean = wineData["pH"].mean()

# Calculate variance

alcoholVariance = wineData["alcohol"].var()
pHVariance = wineData["pH"].var()

print("\nAlcohol Mean:", alcoholMean)
print("Alcohol Variance:", alcoholVariance)

print("\npH Mean:", pHMean)
print("pH Variance:", pHVariance)
```

Output:

```
...    fixed acidity  volatile acidity  citric acid  residual sugar  chlorides  \
    0            7.4              0.70         0.00             1.9      0.076
    1            7.8              0.88         0.00             2.6      0.098
    2            7.8              0.76         0.04             2.3      0.092
    3           11.2              0.28         0.56             1.9      0.075
    4            7.4              0.70         0.00             1.9      0.076

       free sulfur dioxide  total sulfur dioxide  density    pH  sulphates  \
    0                 11.0                  34.0   0.9978  3.51       0.56
    1                 25.0                  67.0   0.9968  3.20       0.68
    2                 15.0                  54.0   0.9970  3.26       0.65
    3                 17.0                  60.0   0.9980  3.16       0.58
    4                 11.0                  34.0   0.9978  3.51       0.56

       alcohol  quality  Id
    0      9.4        5   0
    1      9.8        5   1
    2      9.8        5   2
    3      9.8        6   3
    4      9.4        5   4
    ---------------------------------------------------------------------------

    Alcohol Mean: 10.442111402741325
    Alcohol Variance: 1.171147338035852

    pH Mean: 3.3110148731408575
    pH Variance: 0.02454362762448048
```

Observation:

- The mean alcohol content is approximately **10.42**, with a variance of about **1.17**.
- The mean pH value is approximately **3.31**, with a variance of about **0.025**.
- The higher variance of alcohol indicates that alcohol values have more variation than pH values.